MultiCore TECHNOLOGIES

LEARN TODAY BUILD TOMORROW

Your Future in Tech

CYBER SECURITY COURSE

Build Skills. Stop Threats. Secure Tomorrow.



COURSE HIGHLIGHTS

- Cyber Security Fundamentals
- Network & System Security
- Cryptography & Encryption
- Ethical Hacking & Web Application Security
- Security Operations & Cloud Security

CYBER SECURITY COURSE

Learn • Analyze • Build • Secure

COURSE OVERVIEW

A practical, career-focused Cyber Security program covering security fundamentals, network and system protection, cryptography, ethical hacking, web application security, security operations, incident response, and cloud security.

WHAT WILL YOU LEARN?

CYBER SECURITY FUNDAMENTALS

- Introduction to Cyber Security
- Types of Cyber Threats
- CIA Triad
- Security Policies & Standards
- Cyber Security Frameworks
- Security Roles & Responsibilities

NETWORK SECURITY

- OSI & TCP/IP Models
- Network Protocols
- Firewall Concepts & Types
- VPN
- Secure Network Access
- Network Security Fundamentals

SYSTEM SECURITY

- Operating System Security
- User & Access Management
- Patch Management
- Malware & Antivirus Protection
- Host-Based Intrusion Detection
- Security Hardening
- System Monitoring & Logging

CRYPTOGRAPHY & ENCRYPTION

- Cryptography Fundamentals
- Symmetric & Asymmetric Encryption
- Public Key Infrastructure
- Digital Signatures & Certificates
- Hashing Algorithms
- SSL/TLS
- AES & RSA
- Key Management

ETHICAL HACKING

- Ethical Hacking Concepts
- Footprinting & Reconnaissance
- Scanning & Enumeration
- Vulnerability Assessment

WEB APPLICATION SECURITY

- Web Application Fundamentals
- OWASP Top 10
- SQL Injection
- Cross-Site Scripting
- CSRF
- Authentication & Session Security

SECURITY OPERATIONS

- Security Operations Center
- Incident Response
- Threat Intelligence
- Event Monitoring
- SIEM Tools

CLOUD SECURITY

- Cloud Computing Fundamentals
- Public, Private & Hybrid Cloud
- IaaS, PaaS & SaaS
- Cloud Security Threats
- IAM
- Cloud Data Encryption
- Compliance & Governance



BUILD YOUR CYBER SECURITY SKILLS FOR A SAFER TOMORROW

WHO CAN JOIN?

- Students & graduates
- CSE / IT / ECE students
- Beginners interested in Cyber Security
- Aspiring Cyber Security professionals
- Networking professionals
- System administrators
- IT professionals
- Career changers
- Security enthusiasts

LEARNING OUTCOMES

By the end of the course, learners can:

- Understand fundamental Cyber Security concepts
- Identify common cyber threats and attack surfaces
- Apply the CIA Triad to security scenarios
- Understand network security architecture
- Configure and understand firewall and VPN concepts
- Apply system security and hardening principles
- Understand encryption, hashing and digital signatures
- Perform basic security reconnaissance and vulnerabilty assessment
- Understand common web appplication vulnerabilities
- Analyze security events and logs
- Understand SOC and SIEM operations
- Follow the incident response lifecycle
- Understand cloud security and IAM concepts
- Apply security best practices to real-world scenarios

CYBER SECURITY & NETWORK SECURITY



BUILD MODERN SECURITY SOLUTIONS

DEVELOP SECURE SYSTEMS

STAY AHEAD OF THREATS

GROW YOUR CYBER SECURITY CAREER

SKILLS TODAY FOR A SAFER TOMORROW

CYBER SECURITY FUNDAMENTALS

Understand the Foundation of Information Security

INTRODUCTION TO CYBER SECURITY

- Cyber Security fundamentals
- Information security concepts
- Digital assets and attack surfaces
- Cyber attack lifecycle
- Security objectives
- Security roles & responsibilities

TYPES OF CYBER THREATS

- Malware
- Phishing
- Social engineering
- Password attacks
- Denial-of-Service attacks
- Insider threats
- Data breaches
- Ransomware
- Network-based attacks

CIA TRIAD

CONFIDENTIALITY
Protect sensitive information from unauthorized access.

INTEGRITY
Ensure information remains accurate and protected from unauthorized modification.

AVAILABILITY
Ensure systems and information remain accessible when required.

SECURITY POLICIES & STANDARDS

- Information security policies
- Password policies
- Access control policies
- Acceptable-use policies
- Security standards
- Compliance fundamentals

CYBER SECURITY FRAMEWORKS

- Security framework concepts
- Risk identification
- Protection strategies
- Detection mechanisms
- Response planning
- Recovery concepts

NETWORK SECURITY FUNDAMENTALS

Protect Networks, Devices & Communication

OSI & TCP/IP MODELS

- OSI model
- TCP/IP model
- Network layers
- Network communication
- Security considerations at different layers

NETWORK PROTOCOLS

- TCP/IP
- HTTP / HTTPS
- DNS
- DHCP
- FTP
- SSH
- SMTP
- Network ports & services

FIREWALL CONCEPTS

- Firewall fundamentals
- Network traffic filtering
- Packet filtering
- Stateful firewalls
- Application firewalls
- Firewall rules
- Network segmentation

VPN & SECURE NETWORK ACCESS

- VPN fundamentals
- Secure remote access
- Tunneling concepts
- Encryption in VPN
- Remote-user security
- Secure network communication

KEY SKILLS

Identify → Analyze → Protect → Monitor → Respond

SYSTEM, CRYPTOGRAPHY, ETHICAL HACKING & WEB SECURITY

SYSTEM SECURITY

OPERATING SYSTEM SECURITY
- OS security fundamentals
- Windows security concepts
- Linux security concepts
- System permissions
- Secure configuration

USER & ACCESS MANAGEMENT
- User accounts
- Authentication
- Authorization
- Access control
- Role-based access
- Privilege management

SYSTEM PROTECTION
- Patch management
- Malware protection
- Antivirus solutions
- Host-based intrusion detection
- Security hardening
- Secure system configuration

MONITORING & LOGGING
- System logs
- Security events
- Log collection
- Event monitoring
- Security alerts
- Basic log analysis

CRYPTOGRAPHY & EINCRYPTION

Protect Data Through Secure Communication

CRYPTOGRAPHY FUNDAMENTALS
- Encryption concepts
- Plaintext & ciphertext
- Encryption keys
- Cryptographic algorithms

SYMMETRIC & ASYMMETRIC ENCRYPTION

SYMMETRIC
- Single-key encryption
- AES
- Encryption & decryption

ASYMMETRIC
- Public & private keys
- RSA
- Secure key exchange

PKI & DIGITAL SECURITY
- Public Key Infrastructure
- Digital certificates
- Certificate Authorities
- Digital signatures
- Authentication

HASHING
- Hashing fundamentals
- Password hashing
- Integrity verification
- Hash functions

SSL/TLS
- Secure communication
- TLS fundamentals
- HTTPS
- Certificates
- Secure data transmission

KEY MANAGEMENT
- Key generation
- Key storage
- Key protection
- Key rotation
- Key lifecycle

ETHICAL HACKING & PENETRATION TESTING

Understand Security Testing & Vulnerability Discovery

ETHICAL HACKING CONCEPTS
- Ethical hacking fundamentals
- Security testing
- Scope & authorization
- Rules of engagement
- Vulnerability identification

FOOTPRINTING & RECONNAISSANCE
- Information gathering
- Passive reconnaissance
- Active reconnaissance
- Attack surface identification

SCANNING & ENUMERATION
- Network scanning
- Port scanning concepts
- Service identification
- System enumeration

VULNERABILITY ASSESSMENT
- Vulnerability identification
- Risk assessment
- Security weaknesses
- Vulnerability prioritization
- Remediation concepts

WEB APPLICATION SECURITY

Secure Modern Web Applications

WEB APPLICATION FUNDAMENTALS
- Client-server architecture
- HTTP/HTTPS
- Requests & responses
- Cookies & sessions
- Web application security concepts

OWASP TOP 10
- Understanding common web application risks
- Vulnerability identification
- Secure coding principles
- Risk mitigation

COMMON WEB VULNERABILITIES
- **SQL INJECTION**
  Understanding database query manipulation risks and prevention techniques.
- **CROSS-SITE SCRIPTING — XSS**
  Understanding malicious script injection and input/output security.
- **CSRF**
  Understanding unauthorized actions performed through authenticated sessions.

AUTHENTICATION & SESSION SECURITY
- Secure authentication
- Password security
- Session management
- Access control
- Session protection
- Secure login practices

BUILD MODERN SECURITY SOLUTIONS

DEVELOP SECURE SYSTEMS

STAY AHEAD OF THREATS

GROW YOUR CYBER SECURITY CAREER

SKILLS TODAY FOR A SAFER TOMORROW

KEY HIGHLIGHTS

Understand → Explore → Secure → Test → Defend

SECURITY OPERATIONS, CLOUD SECURITY & CAREER OUTCOME

SECURITY OPERATIONS & INCIDENT RESPONSE

Detect • Analyze • Respond • Recover

SECURITY OPERATIONS CENTER — SOC

- SOC fundamentals
- Security monitoring
- Alert management
- Security event analysis
- Threat detection
- Incident escalation

INCIDENT RESPONSE LIFECYCLE

1. **PREPARATION** — Establish security policies, procedures and response capabilities.
2. **DETECTION & ANALYSIS** — Identify suspicious activity and analyze security events.
3. **CONTAINMENT** — Limit the impact and prevent further spread.
4. **ERADICATION** — Remove the underlying threat.
5. **RECOVERY** — Restore systems and services securely.
6. **LESSONS LEARNED** — Review incidents and improve security controls.

THREAT INTELLIGENCE

- Cyber threat intelligence
- Indicators of compromise
- Threat actors
- Attack patterns
- Threat analysis
- Security awareness

EVENT LOGGING & MONITORING

- Security logs
- Event correlation
- Alert analysis
- Monitoring techniques
- Security dashboards

SIEM TOOLS — INTRODUCTION

- SIEM fundamentals
- Log aggregation
- Event correlation
- Security alerts
- Incident investigation

CLOUD SECURITY

CLOUD COMPUTING

- Cloud computing fundamentals
- Cloud architecture
- Cloud service models

DEPLOYMENT MODELS

- Public Cloud
- Private Cloud
- Hybrid Cloud

SERVICE MODELS

- IaaS — Infrastructure as a Service
- PaaS — Platform as a Service
- SaaS — Software as a Service

CLOUD SECURITY THREATS

- Misconfiguration
- Unauthorized access
- Data exposure
- Account compromise
- Insecure interfaces
- Cloud-based attacks

IDENTITY & ACCESS MANAGEMENT — IAM

- Identity management
- Authentication
- Authorization
- Role-based access
- Least privilege
- Access policies

CLOUD DATA SECURITY

- Data encryption
- Data protection
- Key management
- Secure data storage
- Secure data transmission

COMPLIANCE & GOVERNANCE

- Security governance
- Risk management
- Security policies
- Compliance fundamentals
- Data protection requirements

PRACTICAL LEARNING & SECURITY PROJECT

CYBER SECURITY PROJECT

- Security assessment & defense workflow

Identify → Assess → Secure → Monitor
Respond → Recover

PROJECT COMPONENTS

- Network
- System
- Web application
- Security operations
- Cloud

TOOLS & TECHNOLOGIES

CYBER SECURITY TOOLSET

Networking	TCP/IP • DNS • HTTP/HTTPS • VPN • Firewalls
Security	SIEM • IDS/IPS • Antivirus • Vulnerability Assessment
Cryptography	AES • RSA • Hashing • PKI • SSL/TLS
Web Security	OWASP • SQL Injection • XSS • CSRF
Cloud	IAM • Cloud Encryption • Cloud Security • Governance
Operating Systems	Windows • Linux

FINAL OUTCOME

BECOME SECURITY-READY

Learn → Analyze → Secure → Detect
Respond → Defend

By completing the program, learners build a foundation in:

- Cyber Security fundamentals
- Network security
- System security
- Cryptography
- Ethical hacking concepts
- Web application security
- SOC & SIEM fundamentals
- Incident response
- Cloud security
- Security best practices

YOUR CYBER SECURITY JOURNEY STARTS HERE.

Understand Threats • Protect Systems • Detect Attacks • Respond to Incidents

BUILD MODERN SKILLS

DEVELOP SCALABLE SOLUTIONS

GAIN PRACTICAL SKILLS

BECOME INDUSTRY-READY

LEARN TODAY
SECURE TOMORROW